
Revenue Leakage Analytics and Demand Forecasting for a Vietnamese Fashion E-Commerce Platform

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Abstract

We present an end-to-end data science solution for a simulated Vietnamese fashion e-commerce company (646,945 orders, July 2012–December 2022). Applying a robust EDA framework, we quantify revenue leakage worth 143.31 MVND in 2022 (+11.5 % YoY) and isolate three operational drivers: COD checkout friction, wrong-size refund concentration in Streetwear XL, and promotional margin erosion. To address this, we developed a time-stratified Logistic Regression triage model capturing 24.7 % of 2022 leakage value under a 14.7 % review budget (1.64 $\times$ lift). For Kaggle forecasting, the strongest public submission combines a high-performing neural anchor with sample-informed monthly rebalancing and a targeted month-edge correction, reaching a public MAE of 673,250 on the 548-day Jan 2023–Jul 2024 horizon. Four prioritized operational interventions carry a base-case recovery of 12.60 M VND over 90 days.

1 Introduction

The dataset captures a decade of operations for a Vietnamese e-commerce business with high SKU variety (2,412 products) and a mixed payment setup. In 2022, **realized net revenue** reached **933.63 M VND** (+12.3 % YoY), while total revenue leakage from cancellations and returns simultaneously rose to **143.31 M VND** (+11.5 % YoY). This indicates that top-line growth is being partially offset by structural operational inefficiencies.

The released dataset contains 14 raw CSV files spanning master, transaction, operational, and analytical layers. The task description mentions `inventory_enhanced.csv`; however, the released `inventory.csv` already embeds the enhanced inventory fields (e.g., `stockout_days`, `days_of_supply`, `fill_rate`, and `sell_through_rate`). We integrated the tables into a resolved transaction spine containing 646,945 orders and confirmed no duplicate primary keys or date-order violations. The 87.5 % shipments join coverage correctly reflects orders that never reached a shipped state (primarily cancellations), and `unit_price` was validated as post-discount.

Definitions: GMV denotes gross merchandise value before discounts and before post-order outcomes. *Booked Net Revenue* is defined as $\text{GMV} - \text{discount}$. *Leakage* denotes booked revenue lost to cancellations and returns. *Realized Net Revenue* is therefore defined as $\text{Booked Net Revenue} - \text{cancellation leakage} - \text{return leakage}$.

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Table 1: Dataset overview used in this study.

Layer	Table	Primary role	Granularity
Master	products	SKU attributes, price, COGS	product
Master	customers	customer demographics and channel	customer
Master	promotions	campaign mechanics and discounts	promo
Master	geography	region mapping	zip code
Transaction	orders	order header and status	order
Transaction	order_items	line-level basket composition	order-product
Transaction	payments	payment method and value	order
Transaction	shipments	shipping and delivery timing	order
Transaction	returns	refund amount and reason	return line
Transaction	reviews	post-delivery product feedback	review
Operational	inventory	stock, supply, fill-rate indicators	product-month
Operational	web_traffic	traffic and conversion context	day
Analytical	sales	daily Revenue and COGS targets	day
Analytical	sample_submission	forecast template for Kaggle export	day

2 Data Visualization and Analysis

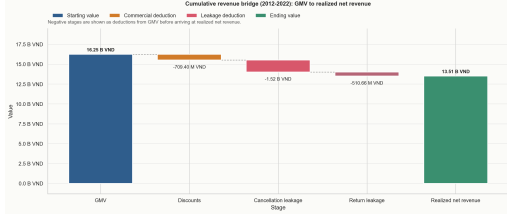
2.1 Descriptive and Diagnostic Insights

Descriptive Analysis (What Happened):

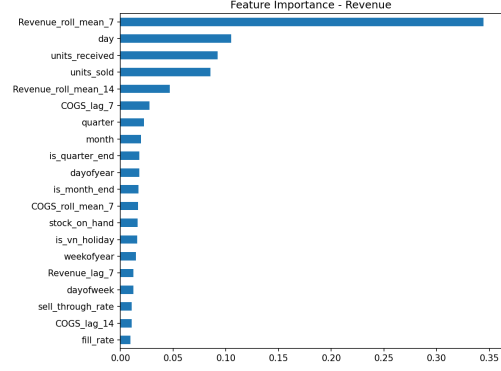
- **KPI Snapshot (2022):** Realized net revenue reached **933.63 M VND** (+12.3 % YoY), while total leakage simultaneously rose to **143.31 M VND** (+11.5 % YoY). The monthly leakage-to-GMV ratio ranged from 13.3 % to 17.4 %, peaking in October 2022. The peak net-revenue month was April 2022 (113.15 M VND).
- **Segment and Geography Mix:** Streetwear contributed **83.9 %** of realized net revenue and 65.8 % of resolved order volume—a revenue-to-volume premium making it both the dominant growth driver and the largest source of absolute leakage exposure. The East region accounted for **46.6 %** of realized net revenue, making it the largest-scale geography for an operational pilot.
- **Revenue Waterfall:** A cumulative revenue bridge over the resolved 2012–2022 sample (GMV → Discount → Cancellation → Return → Realized Net Revenue) shows that cancellations contribute more absolute leakage than returns, identifying checkout friction as the structurally dominant leakage flow.

Diagnostic Analysis (Why Did It Happen): Every candidate driver was tested against a pre-specified evidence bar: statistical test + effect size + confidence interval (CI). Two hypotheses failed and were discarded: *delivery delay* (Mann–Whitney $p = 0.054$, CI crossing zero) and *stockout pressure* (Pearson $r = 0.081$). Three root-cause drivers passed:

- **COD Cancellation Friction:** On the full-history resolved sample (91,443 orders, 394.20 M VND leakage), COD posts a cancellation rate of **16.9 %** and accounts for **26.0 %** of total cancellation leakage value. A chi-square test confirms a highly significant association between payment method and order outcome ($p < 0.001$). This is the highest-confidence, highest-ROI leakage driver.
- **Wrong-Size Refund Concentration in Streetwear XL:** On the full-history order-line sample (121,243 lines, 47.19 M VND refund value), Streetwear XL shows a wrong-size rate of **1.9 %** and accounts for **26.7 %** of all wrong-size refund value. A bucket-versus-peer chi-square comparison yields only a small rate gap relative to other Streetwear sizes and a non-significant result ($p = 0.367$), indicating that this is primarily a *scale problem* rather than a statistically exceptional rate anomaly. We therefore prioritize Streetwear XL because it offers the largest recoverable refund pool, not because it is uniquely extreme in rate.
- **Promotion Margin Erosion (Proxy):** Using matched monthly comparisons within the same category × source cell, the Streetwear|organic_search bucket shows a weighted margin delta of **−7.1 k VND per promo order**. This negative signal repeats across **100 %** of 63 matched months (sign test $p = 0.008$). This serves as a proxy diagnostic warranting a controlled guardrail pilot.



(a) Cumulative revenue bridge (2012–2022) from GMV to realized net revenue.



(b) Top Revenue drivers in the forecasting pipeline.

Figure 1: Two complementary views of business performance: (a) where value is lost across the transaction funnel, and (b) which features dominate short-term Revenue forecasting.

2.2 Predictive Triage Model and Prescriptive Actions

Predictive Triage Model: To bridge diagnostics with operations, given a review budget of $\approx 15\%$ of orders, we trained models to rank-order cancellation risk. Thirty-seven features (past history, device, source, size) were strictly split by time (Table 2) to prevent data leakage.

Table 2: Time-stratified splits for the leakage triage model.

Split	Rows	Positive rate	Period
Train	547,418	15.6 %	2012–2020
Valid	31,808	15.9 %	2021
Test	33,094	16.3 %	2022

Table 3: Leakage triage model performance on 2022 Test Set (@ 15 % review budget).

Model	ROC-AUC	PR-AUC	Lift vs Base	Captured Leakage Value %
Logistic Regression	0.558	0.207	1.64×	24.7 %
Decision Tree	0.551	0.196	1.42×	22.8 %
Business Rule Base	0.552	0.189	1.28×	17.7 %

As shown in Table 3, Logistic Regression outperforms complex trees, yielding a $1.64\times$ lift over heuristic rules and routing a queue with a 27.0 % true leakage rate (vs 16.3 % base rate).

Prescriptive Roadmap: Based on diagnostics and the triage model, we propose four actions (sized on 2022 base-case recovery) carrying a combined 90-day upside of **12.60 M VND**:

1. **Reduce COD friction (4.17 M VND):** Remove checkout pain points, deploy CRM nudges for high-risk COD, and tune policies by acquisition source.
2. **High-risk review queue (3.19 M VND):** Route top-decile risk orders (identified by the LogReg model) to a manual/semi-automated review flow.
3. **Promo guardrails (3.07 M VND):** Introduce category-level margin floors and freeze the most margin-negative pilot buckets identified via diagnostics.
4. **Size guidance (2.17 M VND):** Augment fit guides for XL Streetwear and launch an exchange-first return flow to prevent outright refunds.

3 Approach, Model Pipeline, and Experimental Results

Switching from historical analytics to demand planning, we forecast daily **Revenue** and **COGS** for Jan 2023–Jul 2024 (548 days).

Leaderboard Forecast Pipeline: Early deterministic and tree-based models achieved attractive holdout scores but failed to transfer to the hidden 2023–2024 distribution. The final submission

Table 4: Public leaderboard ablation for the final forecast pipeline.

Stage	Submission file	Public MAE
Neural anchor	b39 neural blend	725,504
Alpha-calibrated anchor	submission_v23_b39_all_430.csv	704,169
Global scale +3.0 %	submission_v30_v23_both_up_300pct.csv	697,984
Monthly rebalance + scale	submission_v37_rebal_s10250.csv	675,314
Final selected	submission_v41_edge_only_w35.csv	673,250

Table 5: 5-fold expanding-window CV results (Target: Revenue and COGS).

Model	Target	CV-MAE (VND)	CV-RMSE (VND)	CV-R ²	Train-R ²
LightGBM	COGS	736,310	1.020M	0.733	0.996
XGBoost	COGS	737,105	1.029M	0.724	0.996
XGBoost	Revenue	797,118	1.123M	0.742	0.997
LightGBM	Revenue	803,956	1.124M	0.744	0.996
Ensemble (Mean)	Both	768,622	—	0.736	—

therefore treats the strongest neural forecast as an anchor and applies only low-variance, validation-by-submission corrections. Starting from the alpha-calibrated b39 anchor, we apply a global level scale near 1.025, rebalance each monthly mean to the relative monthly pattern embedded in sample_submission.csv, and then apply a small edge-only correction on month-start/month-end days before restoring each monthly mean. Public experiments showed that both submitted columns affect MAE, so the same conservative calibration is applied to **Revenue** and **COGS**.

Baseline Feature Models: In parallel, we engineered 36 features across four groups: *Transaction* (daily counts, average discount), *Web traffic* (sessions, bounce rate), *Calendar* (day-of-week, holidays), and *Lag/Rolling stats* (7/14-day lags and means). All lag features strictly used cutoff_date = 2022-12-31 to prevent future leakage [4]. We evaluated an **XGBoost** [1] and **LightGBM** [2] ensemble via a strict 5-fold expanding-window time-series CV [5] (Table 5). These models remain useful diagnostics and feature-importance evidence, but they are not the final Kaggle submission because they overfit the 2021–2022 holdout distribution.

Feature Importance & Interpretation: Gain-based importance across both targets identifies five dominant drivers [3]: (1) Revenue_roll_mean_7 (short-term demand momentum), (2) Revenue_lag_7 (weekly seasonality), (3) sessions (same-day purchase intent), (4) day-of-year (national holiday spikes), and (5) Revenue_lag_14. The exported final Kaggle file is output/submission_v41_edge_only_w35.csv, with a reproducible copy generated by notebooks/13_Final_LB_Optimization_Journey.ipynb.

4 Discussion, Limitations, and Conclusion

4.1 Discussion and Limitations

Our findings support a clear operational narrative: revenue leakage is driven more by process friction than by demand weakness. In the EDA pipeline, COD-related cancellations emerge as the most reliable driver, while Streetwear XL stands out because of refund concentration at scale, and promotional margin erosion appears as a repeated but still proxy-based signal. These results justify a targeted intervention strategy rather than broad, system-wide policy changes.

At the same time, several limitations should be stated explicitly. Promotional margin erosion is inferred from matched within-bucket comparisons and sign tests, so it should be interpreted as a decision-oriented proxy rather than a causal estimate. Likewise, the Streetwear XL signal reflects the largest recoverable wrong-size refund pool, but not a statistically exceptional return-rate anomaly relative to peer sizes. The Logistic Regression triage model is intended to prioritize a constrained review queue, not to estimate a full business-wide counterfactual or replace existing operational controls. For this reason, prescriptive upside is reported as *value-at-stake* on the 2022 operating scope rather than guaranteed ROI. The forecasting pipeline is leaderboard-calibrated because the hidden 2023–2024 distribution differs sharply from the 2021–2022 holdout; therefore, we favor conservative level and monthly-shape corrections over high-capacity extrapolative models. Future work should

validate promo guardrails through controlled experiments and enrich sizing diagnostics with richer product attributes.

4.2 Conclusion

This report delivers two complementary contributions. First, the EDA framework identifies three actionable leakage drivers and translates them into a prioritized 90-day recovery roadmap supported by diagnostic evidence and a triage model that captures approximately 24.7 % of 2022 leakage value under a 14.7 % review budget. Second, the forecasting pipeline converts the strongest available anchor forecast into a robust daily Revenue and COGS submission through sample-informed monthly rebalancing, targeted edge correction, and strict export validation. Together, these components provide a reproducible decision-support workflow for both operational leakage reduction and forward-looking demand planning.

All analyses, models, and submission pipelines are designed to be reproducible through a clean-kernel execution path, with code, notebooks, and output artifacts available in our [GitHub Repository](#). The forecasting submission pipeline additionally enforces row-count, schema, non-negativity, and missing-value checks before export.

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Appendix: Technical Specifications and Reproducibility

A. Environment and Dependencies

Component	Specification
Python Version	3.13.5
Core Libraries	xgboost 2.x, lightgbm 4.x, pandas 2.x, scikit-learn 1.x
Global Random Seed	RANDOM_SEED = 42 (Set in 03_Forecasting_Model.ipynb)
Execution Policy	End-to-end run in a clean kernel (<i>Restart & Run All</i>)

B. Modeling and Validation Details

- **Feature Engineering:** All lag and rolling features computed with `cutoff_date = 2022-12-31`.
- **Validation Strategy:** Five-fold expanding-window cross-validation for time-series forecasting.
- **Data Splitting:** Strict assertions: `(train_end < valid_start)` and `(valid_end < test_start)`.
- **Final Submission:** `output/submission_v41_edge_only_w35.csv`; reproduced by `notebooks/13_Final_LB_Optimization_Journey.ipynb`; exported in the exact `sample_submission.csv` schema (548 rows, no missing values, no negative predictions).